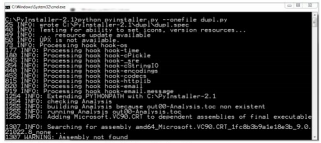

Figure 2: Creating executable file from Python file

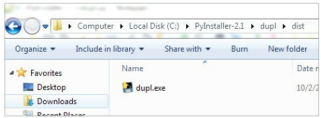

Figure 3: Executable file

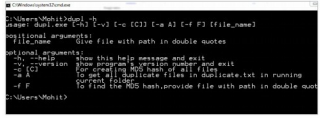

Figure 4: Showing all the options

Figure 5: Creating a pickle file of the hashes of all files

Let us run the program with the following command:

```
1.      dup1 -h
```

This helps to show all options, as seen in Figure 4.

```
2.      dup1 -c 81920
```

The above command will create the database pickle file *mohit.dup1*, which contains the hashes of all files. The argument 81920 is the byte size used by the MD5 algorithm — you can change it. If you increase the size, the calculating speed of the hashes will be increased at the expense of the RAM. If you don't provide the number, then it automatically takes the 4096 byte size.

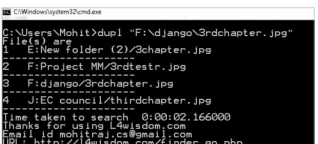

Figure 6: Displaying duplicate files based upon the hash

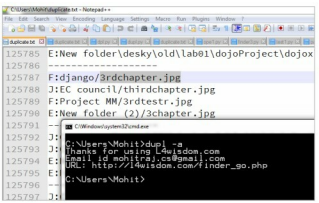

Figure 7: Displaying all the duplicate files in pairs

As you can see in Figure 5, it takes 12 minutes and 53 seconds to create databases. This speed depends upon various factors such as RAM, the model of the computer, the number of files and their sizes. The hash calculation time is directly proportional to the size of the file. The database pickle file will be created in the current running folder.

```
3.      dup1  <file-with full path>
```

Figure 6 shows the duplicate files based upon the hash.

```
4.      dup1 -a
```

In this option, *duplicate.txt* has been created in the current folder, which contains all the duplicate files in pairs, as shown in Figure 7.

```
5.      dup1 -f  <file with path>
```

This option takes a file with path and returns the MD5 hash. END

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